Application Note

BLE_att_server Application Note

Revision History

Date	Version	Description
March 29, 2021	0.50	Initial version created.
April 14, 2021	0.51	Updated format, no content change.
September 29, 2021	0.52	Added 5 Multiple links section.

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1 Overview

This application note describes the settings, functionality, and code flow of the BLE_att_server example code running on an Atmosic-based Bluetooth LE system. This Bluetooth LE system will send advertisements for a Mobile APP connection. When a connection is made, the Mobile APP can send commands using GATT services. Depending on the compile option, this application may enter hibernation after advertisement timeout (default timeout is 180 seconds) if there is no existing connection.

2 Advertisements

The default parameter for advertisement is loaded from Flash NVDS.

	BD address	Device name	Advertisement Interval	Advertisement Timeout
Default value	$USER_BD_ADDR*	Atmosic ATTS	250 ms	180 secs
Flash NVDS Configuration	\tag_data\01-BD_ADDRESS	tag_data\0c-APP_BLE_SCAN_RESP_DATA	\tag_data\06-APP_BLE_ACT_CRT_CMD	tag_data\05-APP_BLE_ACT_STRT_CMD

*__NOTE__: see platform/atmx/ATMx***/examples/README for USER_BD_ADDR usage

After sending an advertisement, you can use Mobile APP to discover the device as shown in Figure 1.

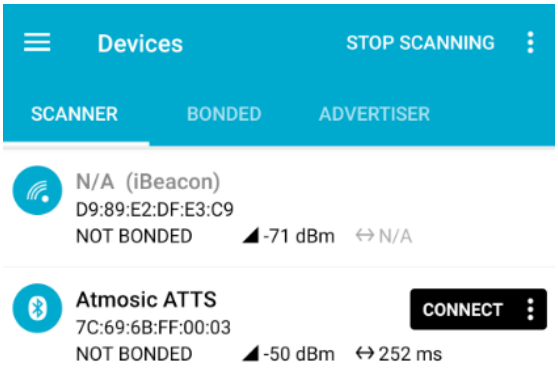


Figure 1 - Device Discovery

3 GATT Service

After the connection with Mobile APP, you will see the “echo write notify service” as shown in Figure 2.

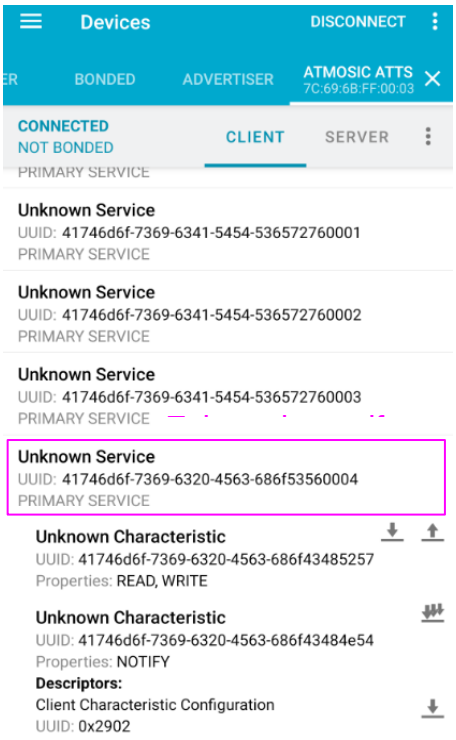


Figure 2 - Echo Write Notify Service

This “echo write notify service” service will echo the write command back to Mobile APP via notification method.

Steps:

a. Enable Notifications in Mobile APP. See Figure 3.

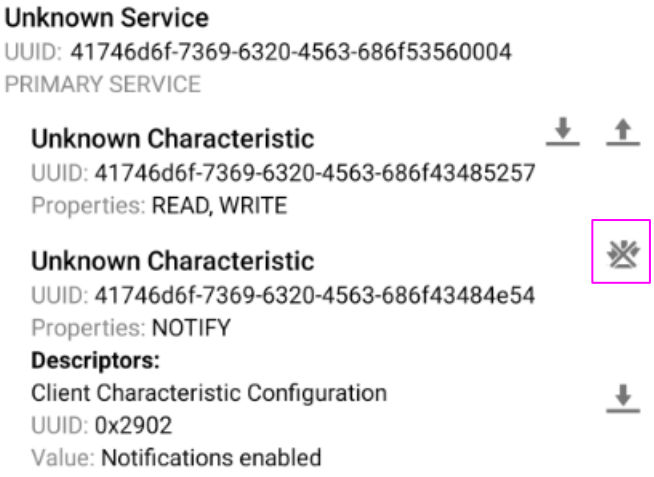


Figure 3 - Enable Notification

b. Send command via GATT write characteristic. See Figure 4.

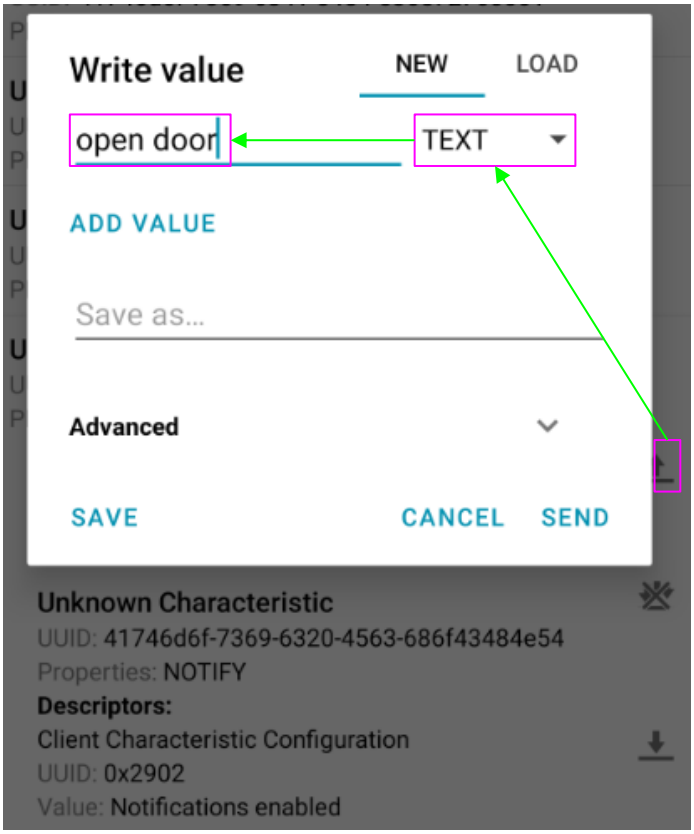


Figure 4 - GATT Write Characteristic

c. Notification shows previous write command. See Figure 5.

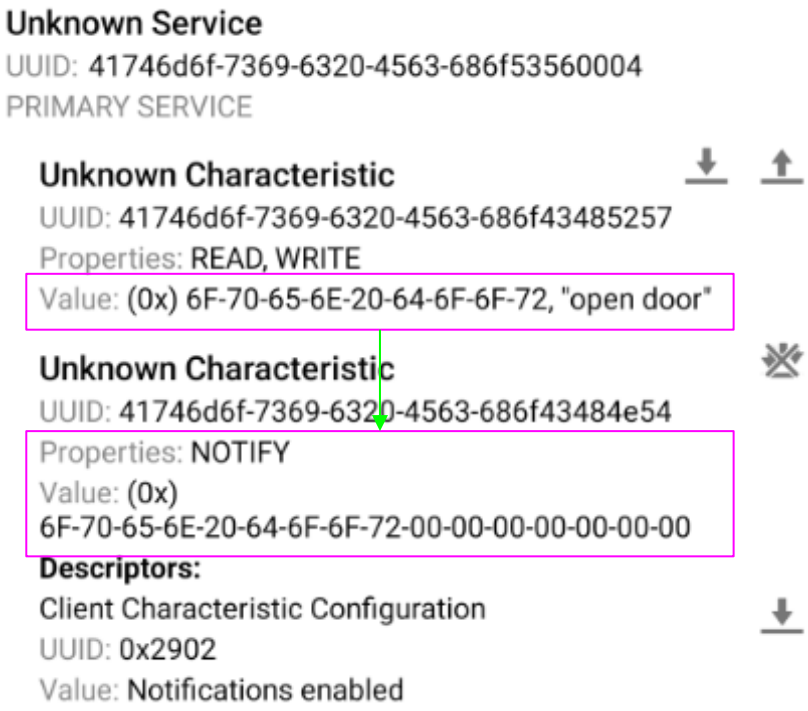


Figure 5 - Previous Write Command Notification

d. If the write command is “open door” or “close door”, the console will display log as shown in Figure 6.

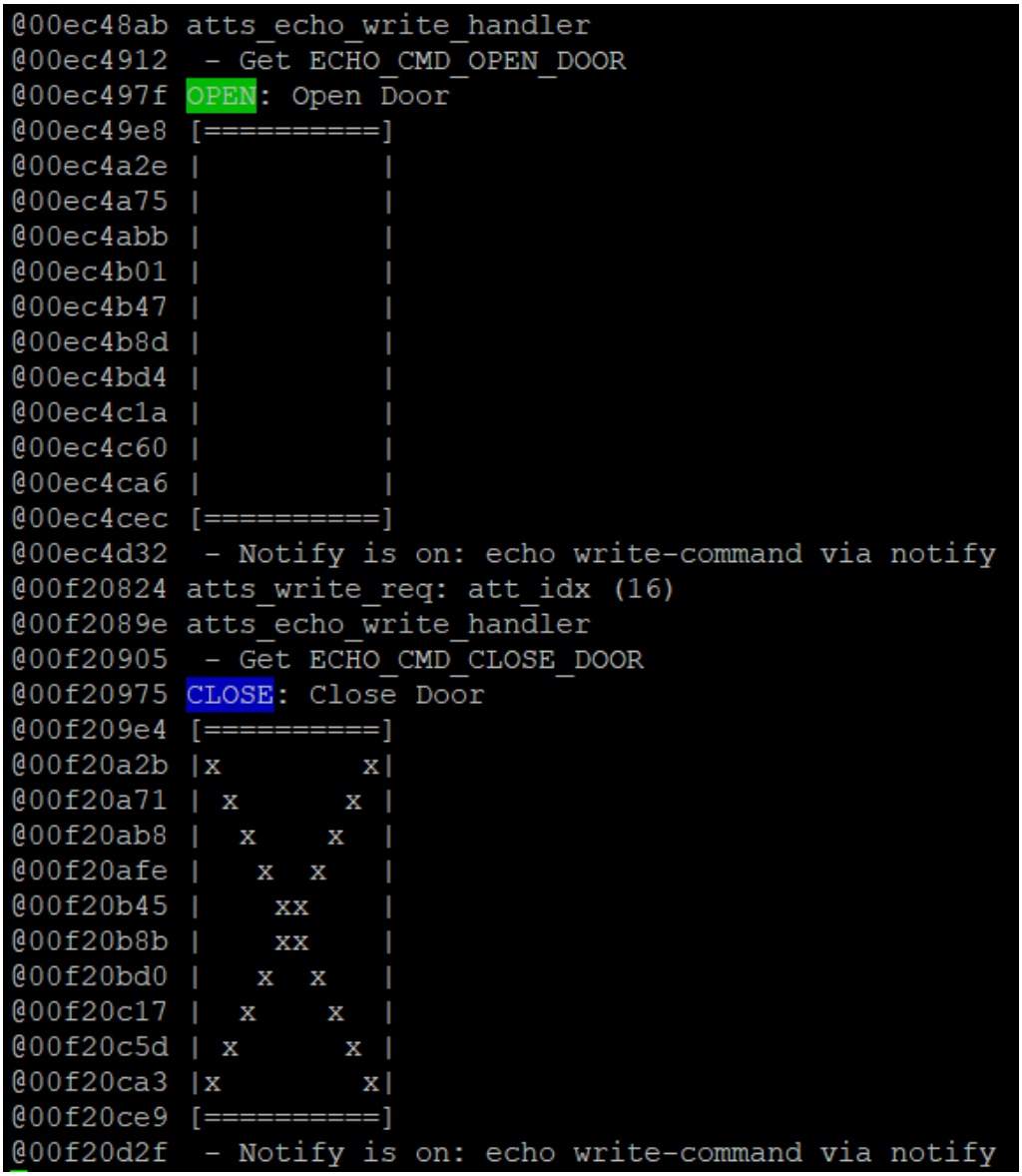


Figure 6 - Open Door and Close Door Write Command Log

4 Create GATT Service

This application uses the Atmosic BLE_ATMPRFS module to create customized GATT services. The first step is registering profile and module initialization.

```
// Register profile
atts_create_prf();
atm_gap_prf_reg(BLE_ATMPRFS, NULL);
```

Below is the example code to create “echo write notify service”. It creates one primary service, two characteristics and one client characteristic configuration. See detail in atts_create_prf() of atts_gatt.c.

```
atts_attr_handle[ATTS_SVC_ECHO] = ble_atmprfs_add_svc(echo_sv_uuid,
        ATTS_SVC_SEC_PROPERTY, &atmprfs_cbs_echo);
atts_attr_handle[ATTS_CHAR_RW_ECHO] = ble_atmprfs_add_char(echo_rw_uuid,
        ATTS_RW_SEC_PROPERTY, ATTS_ECHO_BUF_SIZE);
atts_attr_handle[ATTS_CHAR_NT_ECHO] = ble_atmprfs_add_char(echo_nt_uuid,
        BLE_ATT_NTF_NO_SECURITY, ATTS_ECHO_BUF_SIZE);
atts_attr_handle[ATTS_CHAR_NT_CCCD_ECHO] = ble_atmprfs_add_client_char_cfg();
```

Below is the attribute index after creating “echo write notify service”.

```
enum atts_attr_idx {
    ATTS_SVC_1,
    ATTS_CHAR_RW_1,
    ATTS_CHAR_NT_1,
    ATTS_CHAR_NT_CCCD_1,
    ATTS_SVC_2,
    ATTS_CHAR_IND_2,
    ATTS_CHAR_IND_CCCD_2,
    ATTS_SVC_3,
    ATTS_CHAR_RW_3,
    ATTS_CHAR_RW_CUDD_3,
    ATTS_SVC_ECHO,
    ATTS_CHAR_RW_ECHO,
    ATTS_CHAR_NT_ECHO,
    ATTS_CHAR_NT_CCCD_ECHO,

    ATTS_ATTR_NUM
};
```

Above attribute index will map into the attribute handler. The application will use those handler values to perform ATT related operations in the function call below.

```
static uint8_t atts_read_req(uint8_t conidx, uint8_t att_idx)
{
    ...
    } else if (att_idx == atts_attr_handle[ATTS_CHAR_RW_ECHO]) {
            cfm_size = sizeof(echo_rw_buff);
            cfm = echo_rw_buff;
    } else {
            ASSERT_INFO(0, conidx, att_idx);
    }

    ble_atmprfs_gattc_read_cfm(conidx, att_idx, cfm, cfm_size);

    return ATT_ERR_NO_ERROR;
}


static uint8_t atts_write_req(uint8_t conidx, uint8_t att_idx,
```

```
    const uint8_t *p_data, uint16_t size)
{
    ...
    } else if (att_idx == atts_attr_handle[ATTS_CHAR_RW_ECHO]) {
            memset(echo_rw_buff, 0x00, ATTS_ECHO_BUF_SIZE);

            uint16_t size_write = (size > ATTS_ECHO_BUF_SIZE) ?
               (ATTS_ECHO_BUF_SIZE) : (size);
            memcpy(echo_rw_buff, p_data, size_write);

            // handle echo write command
            atts_echo_write_handler(size_write);
    } else if (att_idx == atts_attr_handle[ATTS_CHAR_NT_CCCD_ECHO]) {
            memcpy(&echo_nt_cccd, p_data, size);
    } else {
            ASSERT_INFO(0, conidx, att_idx);
    }

    return (status);
}
```

The application can use the API below to send notifications to Mobile APP.

```
            ble_atmprfs_gattc_send_ntf(atts_get_curr_conidx(),
            atts_attr_handle[ATTS_CHAR_NT_ECHO],
            echo_rw_buff, ATTS_ECHO_BUF_SIZE, atts_ble_ni_sent);
```

5 Multiple links

This application can support multiple links from Software Development Kit (SDK) 4.2. The ATT_SERVER_LINK_MAX defines the maximal link amount. It can be set from 1 to 5 and it is 5 by default in atts_gatt.h.

```
#ifndef ATT_SERVER_LINK_MAX
#define ATT_SERVER_LINK_MAX 5
#endif
```

It would keep sending advertisements until the connected links reach the ATT_SERVER_LINK_MAX.

For notification and indication, BLE_att_server would send to all connected devices as below:

```
static void atts_send_ntf_ind_all_links(bool is_ntf, uint8_t att_idx,
    uint8_t const *data, uint16_t length)
{
    for (uint8_t link_idx = 0; link_idx < ATT_SERVER_LINK_MAX; link_idx++) {
        if (!atts_status[link_idx].valid) {
            continue;
        }

        uint8_t conidx = atts_status[link_idx].conidx;
        uint8_t status;
        if (is_ntf) {
            status = ble_atmprfs_gattc_send_ntf(conidx, att_idx, data, length,
                atts_ble_ni_sent);
        } else {
            status = ble_atmprfs_gattc_send_ind(conidx, att_idx, data, length,
                atts_ble_ni_sent);
        }

        if (status != ATT_ERR_INSUFF_RESOURCE) {
            continue;
        }

        att_ni_send_req_t *send_reg = malloc(sizeof(att_ni_send_req_t) +
            length - 1);

        ASSERT_ERR(send_reg);

        send_reg->is_ntf = is_ntf;
        send_reg->conidx = conidx;
        send_reg->att_idx = att_idx;
        send_reg->length = length;
        memcpy(&send_reg->data, data, length);
        co_list_push_back(&atts_ble_ni_send_list, &send_reg->hdr);
    }
}
```



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